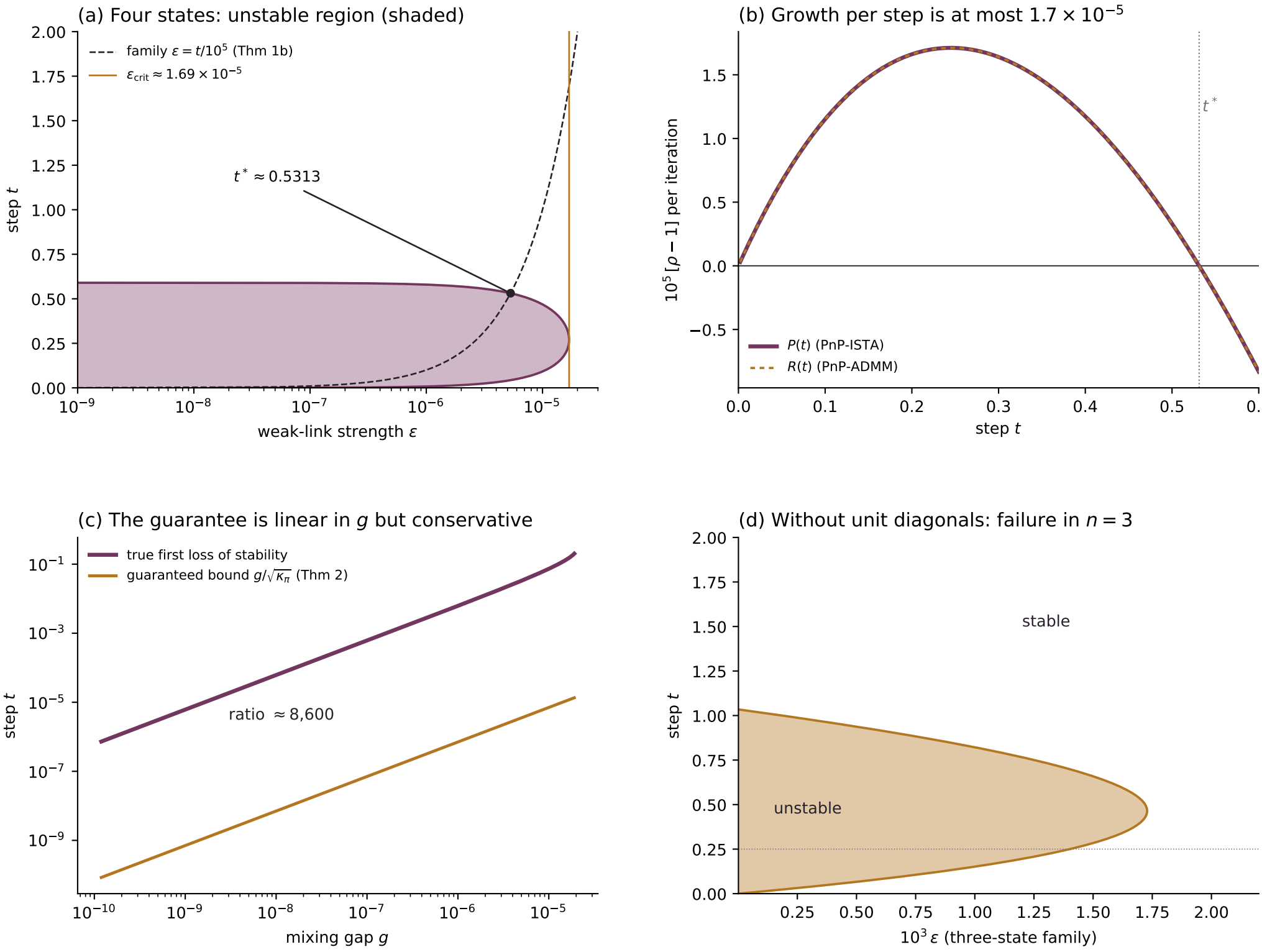
(a) Four states: unstable region (shaded)
family $\varepsilon = t/10^5$ (Thm 1b)
$\varepsilon_{\text{crit}} \approx 1.69 \times 10^{-5}$
$t^* \approx 0.5313$
step t
weak-link strength ε
(b) Growth per step is at most 1.7×10^{-5}
$10^5\,[\rho - 1]$ per iteration
t^*
$P(t)$ (PnP-ISTA)
$R(t)$ (PnP-ADMM)
step t
(c) The guarantee is linear in g but conservative
true first loss of stability
guaranteed bound $g/\sqrt{\kappa_\pi}$ (Thm 2)
ratio $\approx$ 8,600
step t
mixing gap g
(d) Without unit diagonals: failure in $n = 3$
stable
unstable
step t
$10^3\,\varepsilon$ (three-state family)